CONSTRUCTIBILITY CAN BE TESTED AFTER A SURJECTIVE MORPHISM OF QUASI-COMPACT QUASI-SEPARATED SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three parts; the two directions are both written out.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible sheaf of sets, of abelian groups or of modules on the small etale site. Same chapter, the definition of constructibility by a partition into locally closed constructible strata on which the sheaf is locally constant. Supplies the property tested in the three parts of the statement.
- Pullback of a constructible sheaf along a morphism of finite type and the converse for a surjective morphism. Same chapter, the stability of constructibility under pullback, and the criterion producing a partition of Y from one of X along a surjective morphism of quasi-compact quasi-separated schemes. Supplies both directions of the equivalence proved in the proof.

Source reference. Original lemma 095G: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11808–11808. Navigation pointer invoked at proof line 12292; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03SA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12216–12216. Navigation pointer invoked at proof line 12296; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0038: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12298–12298. Navigation pointer invoked at proof line 12298; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095P: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11903–11903. Navigation pointer invoked at proof line 12300; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03Q1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4649–4649. Navigation pointer invoked at proof line 12304; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 12305; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03RZ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11828–11828. Navigation pointer invoked at proof line 12247; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a surjective morphism of quasi-compact and quasi-separated schemes.*

1. *Let $\mathcal{F}$ be a sheaf of sets on $Y_{étale}$. Then $\mathcal{F}$ is constructible if and only if $f^{-1}\mathcal{F}$ is constructible.*
2. *Let $\mathcal{F}$ be an abelian sheaf on $Y_{étale}$. Then $\mathcal{F}$ is constructible if and only if $f^{-1}\mathcal{F}$ is constructible.*
3. *Let Λ be a Noetherian ring. Let $\mathcal{F}$ be sheaf of Λ-modules on $Y_{étale}$. Then $\mathcal{F}$ is constructible if and only if $f^{-1}\mathcal{F}$ is constructible.*

Proof. One implication follows from Lemma 095G. For the converse, assume $f^{-1}\mathcal{F}$ is constructible. Write $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ as a filtered colimit of constructible sheaves (of sets, abelian groups, or modules) using Lemma 03SA. Since f^{-1} is a left adjoint it commutes with colimits (Categories, Lemma 0038) and we see that $f^{-1}\mathcal{F} = \operatorname{colim} f^{-1}\mathcal{F}_i$. By Lemma 095P we see that $f^{-1}\mathcal{F}_i \to f^{-1}\mathcal{F}$ is surjective for all i large enough. Since f is surjective we conclude (by looking at stalks using Lemma 03Q1 and Theorem 03PU) that $\mathcal{F}_i \to \mathcal{F}$ is surjective for all i large enough. Thus $\mathcal{F}$ is the quotient of a constructible sheaf $\mathcal{G}$. Applying the argument once more to $\mathcal{G} \times_{\mathcal{F}} \mathcal{G}$ or the kernel of $\mathcal{G} \to \mathcal{F}$ we conclude using that f^{-1} is exact and that the category of constructible sheaves (of sets, abelian groups, or modules) is preserved under finite (co)limits or (co)kernels inside $Sh(Y_{étale})$, $Sh(X_{étale})$, $Ab(Y_{étale})$, $Ab(X_{étale})$, $Mod(Y_{étale}, \Lambda)$, and $Mod(X_{étale}, \Lambda)$, see Lemma 03RZ. □